

Conceptual Answers: Order-Flow Imbalance

1. Motivation for Measuring OFI at Multiple Depth Levels

- **Richer supply/demand signal.** The best level of the book only captures the tightest quotes. Large or algorithmic traders often split orders across deeper levels, so measuring imbalances at multiple depths reveals “hidden” liquidity pressure that can move prices.
- **Implicit cross-asset pathways.** A best-level order in stock j may propagate into deeper levels of stock i , following paths such as

$$A_j \longrightarrow A_i \longrightarrow \text{OFI}_i^3 \longrightarrow r_i.$$

By sampling OFI at levels $1, \dots, M$, one captures these spillovers without explicitly modeling every cross-asset link.

2. Why LASSO Rather than OLS for Estimating Cross-Impact

- **High dimensionality & multicollinearity.** With $N \approx 100$ candidate cross-asset predictors but only 30 one-minute observations per estimation window, OLS is ill-posed (more regressors than data) and coefficients become unstable.
- **Sparsity assumption.** The authors assume that only a small subset of other stocks meaningfully impacts any given stock’s returns. LASSO naturally zeroes out weak coefficients, yielding a parsimonious set of cross-impact links.
- **Interpretability & stability.** Nonzero LASSO coefficients map directly back to individual tickers’ OFI series. This produces a clear, interpretable network of “neighbors” and guards against overfitting in the presence of collinearity.

3. Why OFI Outperforms Raw Trade Volume for Short-Term Returns

- **Order-book dynamics.** OFI aggregates all limit-order book events—additions, cancellations, executions—so it captures shifts in supply and demand that raw volume (trades only) misses entirely.
- **Directionality.** OFI is a *signed* measure (buys minus sells). A positive OFI indicates net buying pressure (price likely to tick up), while a negative OFI indicates net selling pressure (price likely to drift down). In contrast, trade volume is unsigned and direction-agnostic.